



CHALLENGE 05 / 20

The Silent Ransomware Factory

MALWARE / DIGITAL FORENSICS

DOMAIN	Digital Forensics / Malware Incident Investigation
CASE TYPE	Fictional / Synthetic Student Project
PRIMARY GOAL	Investigate • Correlate • Reconstruct • Report

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Fictional challenge • No real company, person, device or private evidence is required.

01 | CASE SCENARIO

Incident Background

A fictional manufacturing company begins a shift and discovers that a group of business documents can no longer be opened normally. Employees report that the files were available the previous evening, but there is no obvious record showing when the disruption began.

The response team wants to examine application execution, user sessions, file changes and system events around the beginning of the disruption. Several legitimate applications may have been running at the same time, making simple pattern matching insufficient.

The investigation should reconstruct the activity immediately before and during the incident. The final system should help an analyst identify the important sequence and explain which events are connected.

Why the Case Needs Investigation

The organization has enough uncertainty that a simple assumption is not acceptable. Investigators need to combine multiple activity sources, identify the events that belong together, reconstruct the order of activity and document the reasoning behind each important finding.

Investigation principle: Do not treat one record as the complete story. The solution should help an investigator correlate multiple records and clearly distinguish confirmed information from interpretation.

02 | PROBLEM STATEMENT

The Silent Ransomware Factory

Build a digital forensics investigation solution for the fictional case **The Silent Ransomware Factory**. The system should help an investigator organize relevant digital activity, search and filter records, correlate related events, reconstruct a chronological timeline and produce a professional final report. The project is not about obtaining real-world evidence; participants may create synthetic data to demonstrate the workflow.

The solution should be designed as a practical investigator-facing product rather than only a collection of scripts.

03 | WHAT STUDENTS NEED TO BUILD

- Process/application activity
- File change analysis
- User/session analysis
- Timeline
- Event correlation
- Suspicious activity tagging
- Incident report

04 | SUGGESTED INVESTIGATION FLOW

01	CASE CREATION	Create case ID, team/investigator details and case status.
02	DATA INPUT	Import or enter structured synthetic/sample activity data.
03	ANALYSIS	Search, filter and inspect relevant activity.
04	TIMELINE	Arrange important events in chronological order.
05	CORRELATION	Connect related records and identify event relationships.

06	FINDINGS	Record observations and supporting details.
07	REPORT	Generate the final investigation report.

05 | INVESTIGATION QUESTIONS

- What happened before the disruption?
- Which applications were involved?
- When did file changes begin?
- Which files were affected?
- What supports the sequence?

06 | SYNTHETIC DATA FOR DEMONSTRATION

Teams may create fictional records to demonstrate their application. Suggested records can include users/sessions, file activity, application activity, device activity, browser or email metadata, system events and timestamps. No real personal, company or confidential information is required.

DATA CATEGORY	EXAMPLE FIELDS
User / Session	Username, session ID, login/logout time, status
File Activity	Filename, path, action, timestamp, user
Device Activity	Device name, identifier, connection/disconnection time
Application Activity	Application, execution time, user, related file
Browser / Email	URL/message metadata, timestamp, attachment/download
System Events	Event type, timestamp, source, description

07 | EXPECTED FINAL OUTPUT

- **Working Investigation Tool** — functional end-to-end workflow.
- **Case Dashboard** — case information and status.
- **Analysis Views** — organized activity and artifacts.
- **Incident Timeline** — chronological reconstruction.
- **Correlation / Findings** — related events and documented observations.
- **Final Report** — professional investigation summary.
- **Source Code + README** — setup and execution instructions.
- **Live Demo** — clear demonstration to judges.

08 | FINAL REPORT FORMAT

- Case Summary
- Investigation Objective
- Data / Synthetic Inputs Used
- Methodology
- Incident Timeline
- Findings
- Supporting Details / Screenshots
- Conclusion
- Limitations

09 | EVALUATION AREAS

- Understanding of the scenario
- Technical implementation
- Forensic analysis workflow
- Timeline quality
- Event correlation
- Clarity of findings
- User interface/investigator experience
- Final report
- Demo and explanation

10 | CONSTRAINTS

- This is a fictional hackathon case.
- Do not use real people's private or confidential information.
- Synthetic/sample data may be created by participants.
- Do not present synthetic data as evidence from a real incident.
- Conclusions should be based on the data and functionality demonstrated by the project.

11 | CHALLENGE GOAL

Build a digital forensics solution that helps an investigator understand the incident, reconstruct the timeline, correlate relevant activity and produce a professional investigation report.

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